National University of Computer and Emerging Sciences

Islamabad Campus

Statistical Modeling (MT-2002)

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Course Instructors

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Sessional I Exam

Total Time (Hrs): 1

Total Marks: 50

Total Questions: 5

Roll No _____ Section _____ Student Signature _____

Attempt all the questions
Use answer sheet to answer questions 1 - 4
Question 5 should be attempted on bubble sheet attached

DO NOT WRITE BELOW THIS LINE

CLO # 1: Use random variables to represent uncertainty.

Question # 1 8 Marks

Consider the Model$_1$ and the following 4×4 grid of figures, where each panel represents a different Beta distribution as a possible prior for Θ, the probability of success in a coin flip experiment. The subplots are arranged in 4 rows and 4 columns. For modeling purposes, use only the graph located in the **2nd row and 1st column** as the prior distribution for Θ. Using this selected prior, answer the following questions:

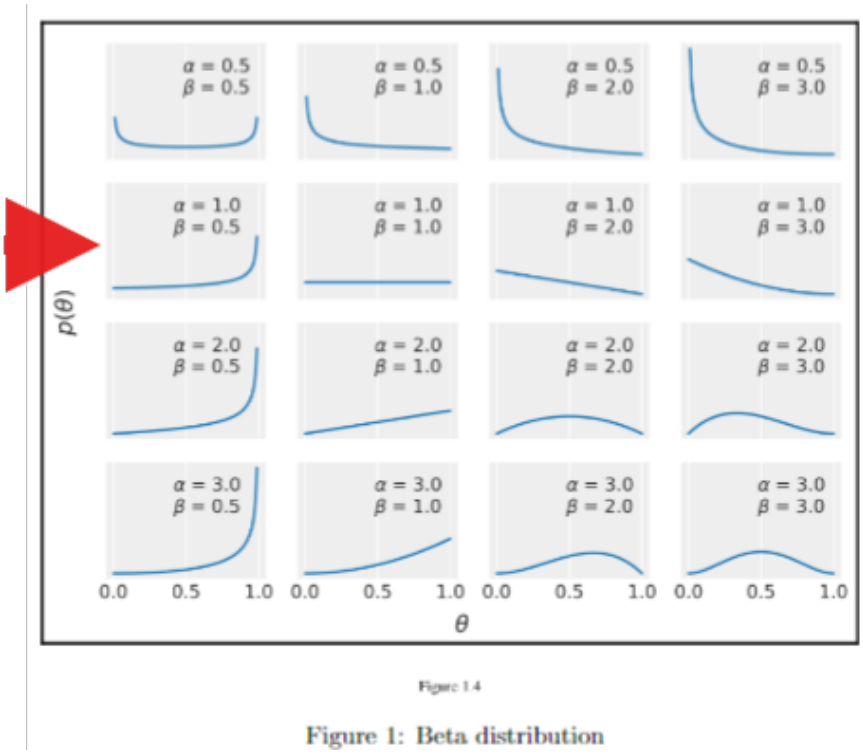

Figure 1: Highlighted Beta distributions as candidate priors

$$\Theta \sim \text{Beta}(\alpha, \beta)$$
$$y \sim \text{Bernoulli}(\Theta)$$

Model$_1$

1. Clearly state the prior and likelihood for the model.

   - **Prior:** The prior for $Model_1$ follows a Beta distribution:

   $$\Theta \sim \text{Beta}(\alpha, \beta) \tag{1}$$

   - **Likelihood:** The likelihood follows a Bernoulli distribution:

   $$y_i \sim \text{Bernoulli}(\Theta) \tag{2}$$

2. Is this a single-parameter or multi-parameter inference problem? What and how many parameter(s) are we inferring from the model? Justify your answer in one line.

   This is a **single-parameter inference problem** because we are estimating only one unknown parameter, Θ. The Beta prior has parameters (α, β), but they are fixed and not inferred from the data.

3. Suppose we are choosing from the highlighted priors, the arrow pointed in the **2nd row** of figure 1. For each of the following scenarios, select a suitable prior and justify your choice:

   (a) If we have **no prior knowledge** about the data.

   Answer: Beta$(1, 1)$.
   Reason: This is a uniform prior, representing no prior knowledge and treating all values of Θ as equally likely.

   (b) If we know that **0s are more frequent than 1s** in the data.

   Answer: Beta$(1, 3)$ Or Beta$(1, 2.0)$.
   Reason: The distribution is skewed toward lower values of Θ, giving higher probability to 0s being more frequent than 1s.

   (c) If we know that **1s are more frequent than 0s** in the data.

   Answer: Beta$(1, 0.5)$.
   Reason: The distribution is skewed toward higher values of Θ, giving higher probability to 1s being more frequent than 0s.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 2 17 Marks

Consider a restaurant operating over a week with daily business statistics:

Day	Mean Revenue (in \$)	Standard Deviation
Monday	1500	160
Tuesday	1200	135
Wednesday	1040	150
Thursday	1080	160
Friday	1240	160
Saturday	1400	230
Sunday	1119	235

Cohens d formula:

$$d = \frac{\mu_2 - \mu_1}{\sqrt{\frac{\sigma_2^2 + \sigma_1^2}{2}}}$$

Compute Cohens d and Categorize the effect sizes based on the computed Cohens d values. Clearly state whether the effect size is small, medium, or large for Monday compared with all other days, and for Friday compared with the two subsequent days. 1 bonus mark will be awarded for a completely correct answer.

Cohen's d Results

Comparison	Mean Difference	Cohen's d	Effect Size
Mon vs Tue	-300	-2.03	Large
Mon vs Wed	-460	-2.97	Large
Mon vs Thu	-420	-2.63	Large
Mon vs Fri	-260	-1.63	Large
Mon vs Sat	-100	-0.51	Medium
Mon vs Sun	-381	-1.90	Large
Fri vs Sat	160	0.81	Large
Fri vs Sun	-121	-0.60	Medium

Table 1: Cohen's d using pooled standard deviation

Note: $d = \frac{\mu_2 - \mu_1}{SD_{pooled}}$. Positive values indicate the second group has a higher mean. Effect size interpretation is based on $|d|$ (Small = 0.2, Medium = 0.5, Large = 0.8).
CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 3 05 Marks

The owner of cafeteria conducted a survey and used Bayesian method to get an estimate of the students who prefer to dine in on Friday. The figure below shows the prior and posterior curves for the proportion of students who prefer to dine in on Friday. What two observations can be made based on this figure?

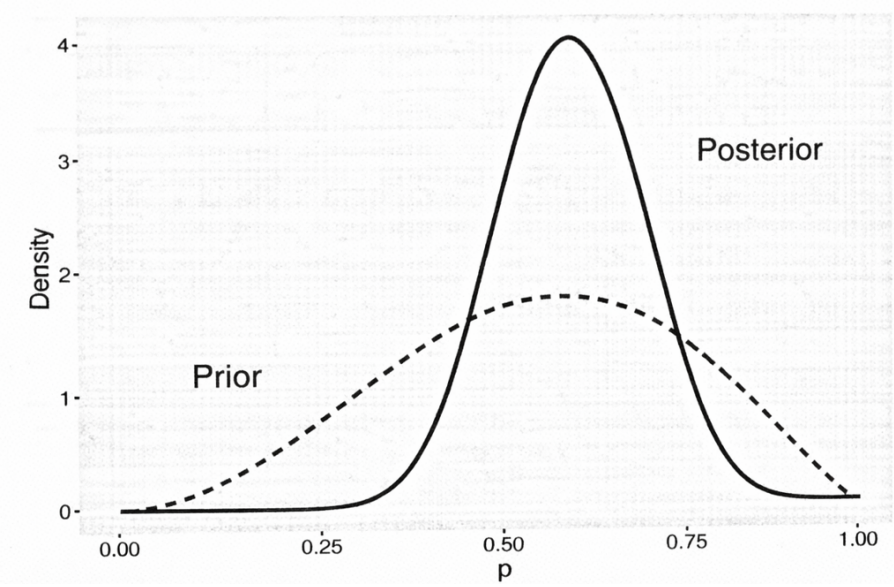

Figure 2: Prior (dashed) and posterior (solid) for proportion p.

Answers

1. The mean value of the prior belief and the posterior is more or less the same.
2. The prior distribution has a much wider spread than the posterior. Initially, the owner was unsure about the proportion of students favoring Friday to dine out. After observing the survey results, the solid posterior curve indicates that he is more certain that p lies between 0.5 and 0.7.

   This highlights a general feature of Bayesian inference: the data sharpens our belief about the parameter of interest, producing a posterior distribution with a smaller spread than the prior distribution.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 4 Based on the following four PyMC models, answer questions next to it in exactly one line. 5 Marks

```
% Model 1
with pm.Model() as model_1:
    prior = pm.Uniform('prior', lower=0, upper=1)
    y = pm.Bernoulli('y', p=prior, observed=data)

    idata_1 = pm.sample(1000)
```

```
% Model 2
with pm.Model() as model_2:
    mu = pm.Uniform('mu', lower=40, upper=70)
    sigma = pm.HalfNormal('sigma', sigma=10)

    y = pm.Normal('y', mu=mu, sigma=sigma, observed=
        data)

    idata_2 = pm.sample(1000)
```

```
% Model 3
with pm.Model() as model_3:
    pm.Normal('normal', mu=0, sigma=1)
    pm.Cauchy('cauchy', alpha=0, beta=1)
    pm.StudentT('studentT', nu=5, mu=0, sigma=1)

    idata_3 = pm.sample_prior_predictive(1000)
```

```
% Model 4
with pm.Model() as model_4:
    mu = pm.Uniform('mu', lower=1, upper=10, shape
        =4)
    sigma = pm.Uniform('sigma', lower=1, upper=10,
        shape=4)

    y = pm.Normal('y', mu=mu[idx], sigma=sigma[idx],
         observed=tip)

    idata_4 = pm.sample(1000)
```

1. What Python construct is used as a context manager to define a PyMC model?

The `with` statement is used as a context manager to define a PyMC model (e.g., `with pm.Model():`).

2. Which keyword argument is used to attach observed data to a likelihood distribution in PyMC?

The `observed` keyword argument is used to attach observed data to a likelihood distribution in PyMC.

3. Out of the above models which function is used to draw posterior samples in PyMC?

The `pm.sample()` function is used to draw posterior samples in PyMC.

4. Out of the above models which function is used to draw prior predictive samples in PyMC?

The `pm.sample_prior_predictive()` function is used to draw prior samples in PyMC.

5. What is the inference variable of model 4

The `idata_4` is the inference variable of model 4.

CLO # 2: Choose appropriate candidate statistical models for a given problem.

Question # 5 15 Marks

1. Which one is a **continuous** probability distribution?

A. Normal
B. Bernoulli
C. Poisson
D. None of the above

2. Which one is a **discrete** probability distribution?

A. Normal
B. Gamma
C. Bernoulli
D. None of the above

3. Which distribution is commonly used for **count data** (0,1,2,3,...)?

A. Poisson
B. Normal
C. Beta
D. None of the above

4. If we are doing inference and inferring a probability p for **two classes** (binary outcome) for a single observation, which likelihood is appropriate?

A. Bernoulli
B. Multinomial
C. Normal
D. None of the above

5. If we are doing inference and inferring a probability vector $\mathbf{p}$ for **more than two classes** for a single observation, which likelihood is appropriate?

A. Categorical
B. Binomial
C. Exponential
D. None of the above

6. If we are doing inference and inferring a probability p for a **specific class** with n trials out of **two classes**, which likelihood is appropriate?

A. Binomial
B. Bernoulli
C. Multinomial
D. None of the above

7. If we are doing inference and inferring a probability vector $\mathbf{p}$ with n trials out of **more than two classes**, which likelihood is appropriate?

A. Multinomial
B. Poisson
C. Categorical
D. None of the above

8. Compared to the Normal distribution, the Student t distribution is generally more:

A. Sensitive to outliers
B. Robust to outliers
C. Skewed to the right
D. None of the above

9. The Student t distribution with $\nu = 1$ is also known as:

A. Cauchy distribution
B. Laplace distribution
C. Uniform distribution
D. None of the above

10. As degrees of freedom ν increases, the Student t distribution becomes closer to:

A. Binomial distribution
B. Exponential distribution
C. Poisson distribution
D. None of the above

11. A **reference value** in Bayesian inference is most commonly:

A. A value of the parameter that represents "no effect" or a meaningful baseline (e.g., 0 for a difference)
B. The maximum value the parameter can take
C. The prior mean of the parameter
D. None of the above

12. In a ROPE-based decision, if a 94% HDI lies **entirely outside** the ROPE around the reference value, a common conclusion is:

A. The parameter is practically equivalent to the reference value
B. There is evidence of a practically meaningful difference from the reference value
C. The posterior must be uniform
D. None of the above

13. Cohen's $d = 0.8$ is most commonly interpreted as:

A. A small effect size
B. A medium effect size
C. A large effect size
D. None of the above

14. If Cohen's d is **positive** and defined as

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}},$$

consider the **Probability of Superiority (PS)**:

$$PS = P(X_1 > X_2).$$

For normally distributed groups,

$$PS = \Phi\left(\frac{d}{\sqrt{2}}\right),$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

If $d > 0$, then

$$PS > 0.50.$$

A. Group 1 is superior
B. Group 2 is superior
C. Both are superior
D. None of the above

15. Which statement correctly distinguishes the **Categorical** and **Multinomial** distributions?

A. Categorical models counts over n trials; Multinomial models a single draw
B. Categorical models a single draw from K classes; Multinomial models counts over n draws from K classes
C. Both are only for two classes
D. None of the above

1	2	3	4	5
A	C	A	A	A
6	7	8	9	10
A	A	B	A	D
11	12	13	14	15
A	B	C	A	B

Table 2: Answer key